succeeded. In a connection-oriented service the first packet sent includes a request for an acknowledgement. If the acknowledgement is received, the sender transmits further packets. If no acknowledgement is received, the sender tries again with the first packet.

17.02 Protocols

The basic definition of a **protocol** is simple – it is a set of rules. So, what do these rules relate to? Before answering, it should be understood that we often talk about 'a protocol' when we are referring to a protocol suite which contains more than one individual protocol. The complexity of networking requires many individual protocols. A further complication is that there might be a number of different versions of a protocol. Finally, there are often protocols that can be used to complement the use of another protocol.

KEY TERM

Protocol: a set of rules for data transmission which are agreed by sender and receiver

The set of rules that constitute a protocol must be agreed between the sender and the receiver for any communication transmitted over a network. At the simplest level, a protocol could define that a positive voltage represents a bit with value 1. A protocol might define a transmission speed that a sender must not exceed. Many of the rules relate to the format of a message or of a component of a message. For example, a definition of the format of the first 40 bytes in a packet.

17.03 A protocol stack

For a protocol suite the protocols can be viewed as layers within a protocol stack. There are a number of aspects relating to this concept.

- Each layer can only accept input from the next higher layer or the next lower layer.
- There is a defined interface between adjacent layers which constitutes the only interaction allowed between layers.
- A layer is serviced by the actions of lower layers.
- With the possible exception of the lowest layer the functioning of a layer is created by installed software.
- A layer may comprise sub-layers.
- Any user interaction will take place using protocols associated with the highest level layer in the stack.
- Any direct access to hardware is confined to the lowest layer in the stack.

17.04 The TCP/IP protocol suite

TCP/IP is the protocol suite underpinning Internet usage. TCP/IP can be explained on the basis of the network model shown in Figure 17.02. It can be seen that TCP/IP only occupies the top three layers of this model.

TIP

Unfortunately there are different names used for two of the layers. Figure 17.02 shows the Network layer and the Data link layer. In some sources these will be named as the Internet layer and the Link layer.

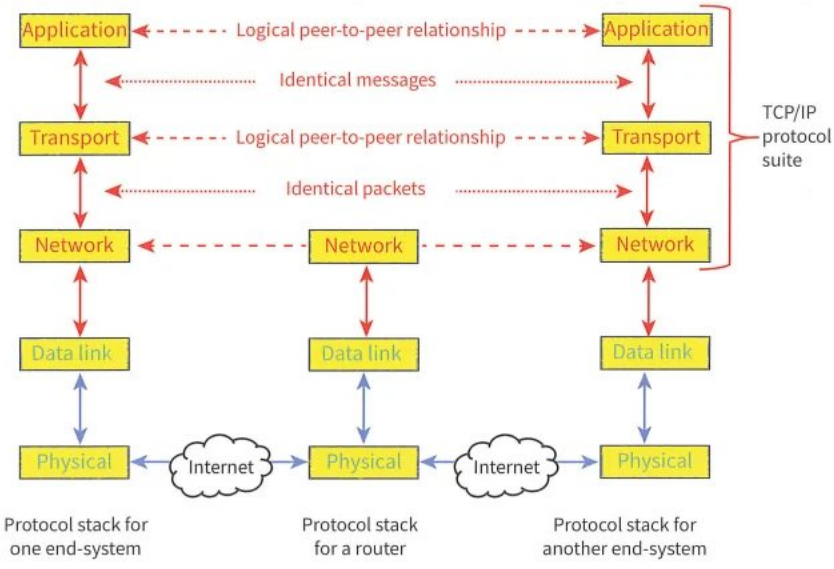


Figure 17.02 A network model relating to the TCP/IP protocol suite

There are two end-systems and the diagram shows the logical connections between corresponding layers in these two systems. An application can run on one end-system and behave as though there was a direct connection with an application running on a different end-system. The application layer protocol on the sender end-system sends a 'message' to the transport layer protocol on the same system. The transport layer protocol then initiates a process which results in the identical 'message' being delivered to the receiver end-system. On the receiver end-system, the final stage is the transport layer protocol delivering the 'message' to the application layer protocol.

The TCP/IP suite comprises a number of protocols, including the following:

- application layer: HTTP, SMTP, DNS, FTP, POP3, IMAP
- transport layer: TCP, UDP, SCTP
- network layer: IP, IGMP, ICMP, ARP.

The selection has been chosen to illustrate that the TCP/IP suite encompasses a very wide range of protocols that is still evolving. Some of the listed protocols will not be considered further.

TCP (Transmission Control Protocol)

If an application is running on an end-system where a 'message' is to be sent to a different end-system the application will be controlled by an application layer protocol as described above. The protocol will transmit the user data to the transport layer. The TCP protocol operating in the transport layer now has to take responsibility for ensuring the safe delivery of the 'message' to the receiver. The TCP protocol creates sufficient packets to hold all of the data. Each packet consists of a header plus the user data.

Question 17.01

Can you find out some details about the format of a packet that TCP would send to IP?

As well as ensuring safe delivery, TCP has to ensure that any response is directed back to the application protocol. So, one item in the header is the port number which identifies the application layer protocol. For example, for HTTP the port number is 80. The packet must also include the port number for the application layer protocol at the receiving end-system. However, TCP is not concerned with the address of the receiving end-system. If the packet is one of a sequence, a sequence number is included to ensure eventual correct reassembly of the user data.

The TCP protocol is connection-oriented. In accordance with the procedure described in Section 17.01, initially just one packet of a sequence is sent to the network layer. Once the network layer returns an acknowledgement to the Transport layer indicating that the connection has been established, TCP sends the other packets and receives response packets containing acknowledgements. This allows missing packets to be identified and re-sent.

IP (Internet Protocol)

The function of the network layer, and in particular of the IP, is to ensure correct routing over the Internet. To do this the IP protocol takes the packet received from the transport layer and adds a further header. The header contains the IP addresses of both the sender and the receiver. To find the IP address of the receiver, it is very likely to use the DNS service to find the address corresponding to the URL supplied in the user data. The DNS service is discussed in some detail in Chapter 2 (Section 2.09).

The IP packet, which is usually called a 'datagram', is sent to the data-link layer and therefore to a different protocol suite. The data-link layer assembles datagrams into 'frames' as discussed in the next section of this chapter. Once the IP packet has been sent to the data-link layer, IP has no further duty. IP functions as a connectionless service. Once a packet has been sent, IP will have no knowledge of whether or not it reached its destination. If IP receives a packet which contains an acknowledgement of a previously sent packet, it will simply pass the packet on to TCP with no awareness of the content.

The router

As Figure 17.02 shows, the frame sent by the data-link layer will arrive at a router during transmission (more likely at several routers). At this stage, the datagram content of the frame is given back to IP. It is now the function of the router software to choose the next target host in the transmission. The software has access to a routing table appropriate to that router. The size and complexity of the Internet prohibits a router from having a global routing table. Once the appropriate address has been inserted into the datagram, IP passes it back to the data-link layer of the router.

The routing table for every router has details of any current problems with any of the options for the next transmission step. This ensures that packets are delivered to their destination in the shortest possible time available.

The major distinction between a switch and a router as a node in a network is that when a frame arrives at a switch, it is transmitted on without any routing decision. A switch operates in the data-link layer but has no access to the network layer.

17.05 The Ethernet protocol stack

As discussed in Chapter 2 (Section 2.05), Ethernet is a protocol suite designed for use in a local area network (LAN). As such it can function in an isolated LAN with no connection to the Internet or any other network. However, it is now almost inevitable that a LAN will be connected to the Internet and, therefore, a LAN's protocol suite will support the protocol suite in use for the Internet.

If we look at the protocol stack for one end-system, as shown in Figure 17.02, we can see that the TCP/IP protocol suite occupies the top three layers of the five-layer stack and is therefore supported by the lower two layers. TCP/IP has no concern with the functioning of these two lower layers; it is designed to be capable of being supported by whatever protocols are available. It should be noted that some sources only use a four-layer stack for an end-system. This is either a decision to only include layers that are handled entirely by software. Or it is a decision to amalgamate all of the support for TCP/IP into one layer.

Ethernet is the most likely protocol to be used to provide the functionality required of the two lower layers. Logically the Ethernet suite can be viewed as comprising two sub-layers for each of the Data link and Physical layers. This is illustrated in Figure 17.03.

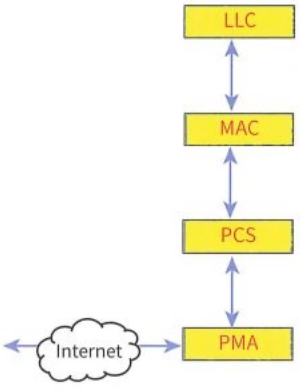


Figure 17.03 The sub-layers of Ethernet

The following points explain how Ethernet functions when supporting TCP/IP.

- The Logical Link Control (LCC) protocol is responsible for the interaction with the Network layer. It manages data transmissions and ensures the integrity of data transmissions. However, because Ethernet is a connectionless protocol it has no responsibility for checking that a transmission has been successfully delivered.
- The Medium Access Control (MAC) protocol is responsible for assembling the Ethernet packet which is referred to as a frame. Two components of this are the address of the transmitter and the address of the receiver. In addition the MAC protocol initiates frame transmission and handles recovery from transmission failure due to a collision (possibly using CSMA/CD).
- The Physical Coding Sublayer (PCS) protocol is responsible for coding data ready for transmission or decoding data received. It either receives a frame from the MAC protocol or sends one to it.
- The Physical Medium Attachment (PMA) protocol is responsible for signal transmitting and receiving.

MAC addresses

Both addresses in the Ethernet frame are examples of what are called physical or MAC addresses. A MAC address is the one which uniquely defines one NIC, as described in Chapter 2 (Section 2.04).

The reason that unique addresses have so far been guaranteed is that the 48 bits used for the definition have given a sufficient number of different addresses. However, there is a 64-bit alternative which is already used occasionally but is available for future use when 48 bits are insufficient. The 48-bit address is usually written in hexadecimal notation, for example:

4A:30:12:24:1A:10

In one version of this addressing scheme the first three bytes identify the manufacturer.

17.06 Application-layer protocols associated with TCP/IP

There are very many application-layer protocols. This discussion considers some that are used most often.

HTTP (HyperText Transfer Protocol)



TIP

Be careful not to confuse HTTP and HTML.

Because HTTP (HyperText Transfer Protocol) underpins the World Wide Web it has to be considered to be the most important application-layer protocol. Every time a user accesses a website using a browser, HTTP is used but its functionality is hidden from view.

HTTP is a transaction-oriented, client–server protocol. The transaction involves the client sending a 'request' message and the server sending back a 'response' message. The HTTP protocol defines the format of the message. The first line of a request message is the 'request line'. Optionally this can be followed by header lines. All of this uses ASCII coding. The format of the request line is:

```
<Method> <URL> <Version>CRLF
```

where CR and LF are the ASCII carriage return and line feed characters. The request line usually has GET as the method. However, there are several alternatives to the GET method which makes HTTP potentially a more widely applicable protocol than just being used for webpage access. The version has to be specified because HTTP has evolved so there is more than one version in use.

Email protocols

The traditional method of sending and receiving emails is illustrated in Figure 17.04 which shows that three, individual, client–server interactions are involved. The email sender acting as a client has a connection to a mail server. This server then has to function as a client in the transmission to the mail server used by the email receiver acting as a client.